Software Requirements Specification (SRS) - Online Quiz System

Project: Online Quiz System (console-based MCQ quiz, C++17)
Version: 2.0
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Date: 06-11-2026
Status: Final for review

Revision history

Version	Date	Author	Change summary	Approval
1.0	28-09-2026	Team	Initial SRS: OQ-F-001 to 016, OQ-NF-001 to 005, OQ-SR-001 to 005	
1.1	28-09-2026	Team	Baseline for Sprint 1 (v1.0 release, 24-10-2026)	
1.2	03-11-2026	Team	Added topic bank and randomised quiz: OQ-F-017 to OQ-F-022	
2.0	06-11-2026	Team	Review update: aligned with the implementation (commit 05bc7d4); restored template structure, acceptance criteria and test references; added security objectives and use-case diagrams; RTM with results	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Guide			
Team Lead			

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1. Introduction

1.1 Purpose

This document specifies the functional, non-functional and security requirements of the Online Quiz System. It is the baseline for the Software Architecture and Design Specification (SAD) v2.0 and the Software Test Plan (STP) v2.0, which use the requirement IDs defined here.

1.2 Scope

The Online Quiz System is a single-user C++17 console application. A user chooses one of five topics, answers five randomly selected multiple-choice questions with immediate feedback, sees the final score and can retry the topic, choose another topic or return to the main menu. An administrator logs in to add, modify, delete and view questions in a local question bank.

Out of scope: graphical, web or mobile interface; network or multi-user use; user accounts for players; storing results or history.

1.3 Audience

Developers, testers, the course instructor, project evaluators and future maintainers.

1.4 Definitions

Term	Definition
MCQ	Multiple Choice Question
Attempt	One run of five questions on one topic, ending with the final score
Question bank	All questions in data/questions.txt
Topic	ML (Machine Learning), DBMS, SE (Software Engineering), DSA (Data Structures and Algorithms), GK (General Knowledge)
Admin	Administrator who maintains the question bank
SAD / STP / RTM	Software Architecture and Design Specification / Software Test Plan / Requirements Traceability Matrix

2. Overall description

2.1 Product perspective

A standalone console program with no server, database or network connection. It reads and writes one local text file, data/questions.txt. It is an academic project developed in sprints; Sprint 1 was released as v1.0.

2.2 Major product functions

- Main menu: 1. Admin login, 2. Take Quiz, 0. Exit.
- Topic selection: ML, DBMS, SE, DSA, GK (0 returns to the main menu).
- Five distinct random questions per attempt from the chosen topic, one at a time, options A-D.
- Answer validation, answer checking with immediate feedback, scoring and final score.
- Result menu: retry the topic (new set, score reset), return to topic selection or the main menu.

- Administrator login with masked password; add, modify, delete (with confirmation) and view questions.
- Validation of question records when loading and before saving.

2.3 User roles and characteristics

- **User (player):** any student with basic keyboard skills; takes quizzes.
- **Administrator:** a team member or instructor who maintains the question bank; logs in first.

2.4 Operating environment

macOS, Linux or Windows terminal on a standard PC; program built with a C++17 compiler (clang++, g++ or MinGW g++) and started from the project folder so that data/questions.txt is found.

2.5 Constraints

- Console and keyboard only.
- C++17 standard library only (password masking uses termios on macOS/Linux and _getch on Windows).
- Question data stored in one local delimited text file.
- Each topic needs at least five valid questions for an attempt.

2.6 Assumptions and dependencies

- The question bank is prepared by the administrator (15 questions per topic are delivered).
- The development build uses a single test administrator account (see 5.1.3).

3. External interface requirements

3.1 User interfaces

- Numbered text menus (main menu, topic menu, result menu, admin menu); every prompt states the valid input.
- Question screen: "Question i of 5: <text>", options A-D, prompt "Your answer (A-D):".
- Feedback after each answer: "Correct!" or "Incorrect!" plus the correct option.
- Result: "Your score: N / 5".

3.2 Hardware interfaces

Standard keyboard and display; no special hardware.

3.3 Software interfaces

- C++17 compiler and standard library.
- Local file system: data/questions.txt (read; rewritten by admin functions).

3.4 Communications

None. The system runs offline.

3.5 Question record format

One question per line: TOPIC|Question text|Option A|Option B|Option C|Option D|Answer, for example:

DSA|Which data structure follows FIFO order?|Stack|Queue|Heap|Graph|B

4. System features (detailed)

Each requirement has acceptance criteria and a test case from the STP. IDs follow OQ-F-###.

4.1 Quiz Operation

Description: The user starts a quiz, answers questions one at a time with input validation, and moves between attempts.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-001	The system shall start a quiz when the user selects "2. Take Quiz" from the main menu.	Functional	High	User	Selecting 2 shows the topic menu; after a topic is chosen, "Question 1 of 5" is shown with the score at 0. Test: TC-01	-
OQ-F-002	The system shall display one multiple-choice question at a time.	Functional	High	User	Only Question 1 is shown first; Question 2 appears only after Question 1 is answered. Test: TC-02	-
OQ-F-003	The system shall display four answer options labelled A, B, C and D for each question.	Functional	High	User	Every question shows exactly four options labelled A-D. Test: TC-03	-
OQ-F-004	The system shall accept a single option letter A-D (upper or lower case) as the answer to the current question.	Functional	High	User	"B" and "b" are both accepted as option B. Test: TC-04	-
OQ-F-005	The system shall reject any other input, show "Invalid option. Enter A, B, C, or D." and ask again for the same question.	Functional	High	User	E, 5, @, AB and empty input are each rejected; the question stays active and the score is unchanged. Test: TC-05	-
OQ-F-009	The system shall move to the next question only after a valid answer has been processed.	Functional	High	User	After an invalid input the same question stays; after a valid one the next question appears. Test: TC-09	-
OQ-F-016	The system shall reset the score to zero at the start of every new attempt.	Functional	Medium	User	After a completed attempt, a retry starts at Question 1 with score 0. Test: TC-16	Navigation choices: OQ-F-022
OQ-F-022	The system shall offer, after the result, "1. Retry this topic", "2. Return to topic selection" and "0. Return to main menu".	Functional	High	User	Each of the three choices goes to the correct screen. Test: TC-32	-

4.2 Answer Checking and Scoring

Description: Each answer is checked, feedback is shown and the score is updated and displayed.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-006	The system shall compare the submitted answer with the correct answer and tell the user whether it was correct, showing the correct option.	Functional	High	User	A correct answer shows "Correct!", a wrong one "Incorrect!"; both show "Correct answer: <letter>. <text>". Test: TC-06	Feedback after each answer (correct option shown)
OQ-F-007	The system shall increase the score by exactly one for each correct answer.	Functional	High	User	Score goes from 0 to 1 after one correct answer. Test: TC-07	-
OQ-F-008	The system shall leave the score unchanged for an incorrect answer.	Functional	High	User	A wrong answer leaves the score at its previous value. Test: TC-08	-
OQ-F-010	The system shall calculate the final score after all five questions of the attempt are answered.	Functional	High	User	With 3 of 5 correct the final score is 3. Test: TC-10	-
OQ-F-011	The system shall display the final score as "Your score: N / 5".	Functional	High	User	Result screen shows "Your score: 3 / 5" for 3 correct answers. Test: TC-11	-

4.3 Topic Question Bank and Randomisation

Description: Questions are stored by topic; each attempt draws five random questions from the chosen topic.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-017	The system shall keep a local question bank organised into the topics ML, DBMS, SE, DSA and GK.	Functional	High	Instructor	questions.txt holds 15 questions for each of the five topics (75 in total). Test: TC-27	Delivered bank: 75 questions
OQ-F-018	The system shall load only records of the form TOPIC|question|A|B|C|D|answer with a valid topic, non-empty text and options and an answer A-D, and shall skip other records with a warning naming the line number.	Functional	High	Developer	Malformed records (wrong field count, unknown topic, answer E, empty text) are skipped with a warning; valid ones are loaded. Test: TC-28	-
OQ-F-019	The system shall let the user choose a topic (1-5) or return to the main menu (0), reject any other choice, and use only questions of the chosen topic.	Functional	High	User	Every question in an attempt belongs to the chosen topic; "9" or "abc" gives "Invalid topic selection." and the menu is shown again. Test: TC-29	-
OQ-F-020	The system shall select five distinct questions at random from the chosen topic, in random order, and shall not start an attempt if fewer than five valid questions exist.	Functional	High	User	Each attempt has exactly five distinct questions; with four valid questions the message "Not enough valid questions for topic ..." is shown. Test: TC-30	-

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-021	The system shall avoid giving the same five-question set on two consecutive attempts of a topic when another set is possible.	Functional	Medium	User	On retry the complete set differs from the previous attempt; the question bank itself is not changed. Test: TC-31	Needs more than 5 questions in the topic

4.4 Question Management and Access

Description: An authenticated administrator adds, modifies, deletes and views questions.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OQ-F-012	The system shall allow an authenticated administrator to add a question with its topic.	Functional	High	Administrator	Added question is saved to questions.txt with its topic and appears in a quiz on that topic. Test: TC-12	-
OQ-F-013	The system shall allow an authenticated administrator to modify an existing question.	Functional	High	Administrator	Updated text and answer are saved and used in the next quiz. Test: TC-13	-
OQ-F-014	The system shall allow an authenticated administrator to delete a selected question after Y/N confirmation.	Functional	High	Administrator	After "Y" the question is removed from questions.txt; after "N" nothing changes. Test: TC-14	Admin can also view all questions
OQ-F-015	The system shall prevent unauthenticated users from reaching any question-management function.	Functional	High	Administrator / Security	Add, modify, delete and view are reachable only from the admin menu after login; other input gives "Invalid menu choice." and no data changes. Test: TC-15	See OQ-SR-001, OQ-SR-004

5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	Acceptance criteria / Test case ref
OQ-NF-001	The system should process normal user input and show the next screen within 1 second.	Performance	Medium	At least 95% of 20 measured inputs are processed within 1 second. Test: TC-17
OQ-NF-002	The system should give the same score for the same answers and keep the score correct throughout an attempt.	Reliability	High	The same answer script gives the same score on 3 runs. Test: TC-18
OQ-NF-003	Console prompts and messages should state what input is expected.	Usability	Medium	3 new users complete a quiz without extra help. Test: TC-19
OQ-NF-004	The normal user flow should not expose administrator functions or reveal a correct answer before the user has answered.	Security	High	No correct answer appears before an answer is submitted; no management option appears in the user menus. Test: TC-20

Req ID	Requirement	Category	Priority	Acceptance criteria / Test case ref
OQ-NF-005	The program should build with a C++17 compiler (clang++, g++ or MinGW g++) with no warnings and run without special hardware.	Portability	Medium	Clean build (-Wall -Wextra) and a full quiz run on macOS, Linux and Windows. Test: TC-21

5.1 Security

5.1.1 Security Objectives

ID	Objective	Description
SO-1	Question bank integrity	Only an authenticated administrator can change the question bank, and only valid records are saved.
SO-2	Credential confidentiality	Administrator credentials are never displayed while typing or in any program output.
SO-3	Separation of user and admin functions	A normal user cannot reach administrator functions or see answers before answering.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref	Objective
OQ-SR-001	The system shall require a successful administrator login before any question-management function is available.	Security	High	Choosing "1. Admin login" always shows the login prompt first; the admin menu appears only after a correct login. Test: TC-22	SO-1, SO-3
OQ-SR-002	The system shall reject wrong or empty administrator credentials.	Security	High	Wrong username or password gives "Access denied. Invalid username or password."; empty fields give "Username and password cannot be empty." Test: TC-23	SO-1
OQ-SR-003	The system shall validate administrator input (valid topic, non-empty text and four options, answer A-D, no "|" character) before saving question data.	Security	Medium	Empty text, a missing option or answer E are rejected and questions.txt is unchanged. Test: TC-24	SO-1
OQ-SR-004	The system shall prevent normal users from opening administrator functions directly from any user menu or prompt.	Security	High	Inputs such as admin, 9, 99 and -1 are rejected at the main menu; menu numbers typed at an answer prompt are rejected as answers. Test: TC-25	SO-3
OQ-SR-005	The system shall mask the password with "*" while it is typed and never display administrator credentials in console output.	Security	Medium	Password shows as "*" in a terminal; the password never appears in any output. Test: TC-26	SO-2

5.1.3 Known limitation

The development build uses one test account (admin / admin123) defined in src/main.cpp. This meets OQ-SR-001 to OQ-SR-005 for the course demo, but the credentials must be replaced (for example by a hashed credential file) before any real use.

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: all High-priority functional and security requirements implemented and verified, no open Critical or High defect, and the RTM (section 8) showing a recorded result for every requirement.

Acceptance test suites (test cases in STP section 13.2):

Suite	Covers	Test cases
Menu and Navigation	Main menu, quiz start, topic menu, result menu, exit	TC-01, TC-29, TC-32
Quiz Delivery and Input Validation	One question at a time, options A-D, valid/invalid answers, next question	TC-02 to TC-05, TC-09
Answer Checking and Scoring	Correct/incorrect, score update, final score, reset	TC-06 to TC-08, TC-10, TC-11, TC-16
Topic Bank and Randomisation	Topic contents, record validation, 5 random questions, new set on retry	TC-27, TC-28, TC-30, TC-31
Question Management	Add, modify, delete	TC-12 to TC-14
Administrator Security	Access control, login, credentials, input validation	TC-15, TC-20, TC-22 to TC-26
Quality and Portability	Response time, reliability, usability, build on 3 platforms	TC-17 to TC-19, TC-21

7. System models and diagrams

7.1 UML Use-Case diagrams

UC Diagram 1 - Player. Taking a quiz includes topic selection, which includes drawing five random questions from the topic. Answering includes input validation and answer checking with feedback. Retry draws a new set; returning to topic selection shows the topic menu again.

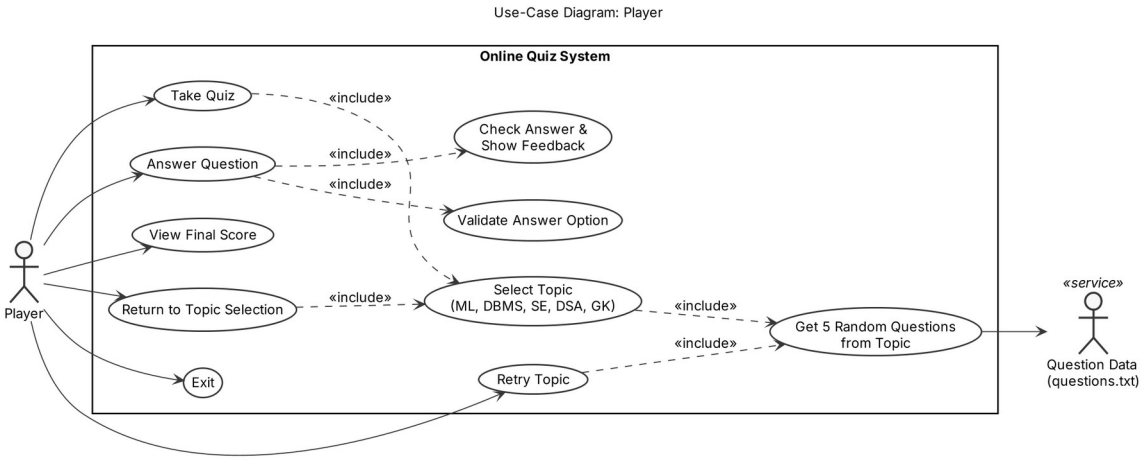


Figure 7.1 - Use-case diagram: Player

UC Diagram 2 - Quiz Administrator. Login includes masked password entry and credential checking. Add and modify include topic selection and validation; delete includes Y/N confirmation; all three save the question bank.

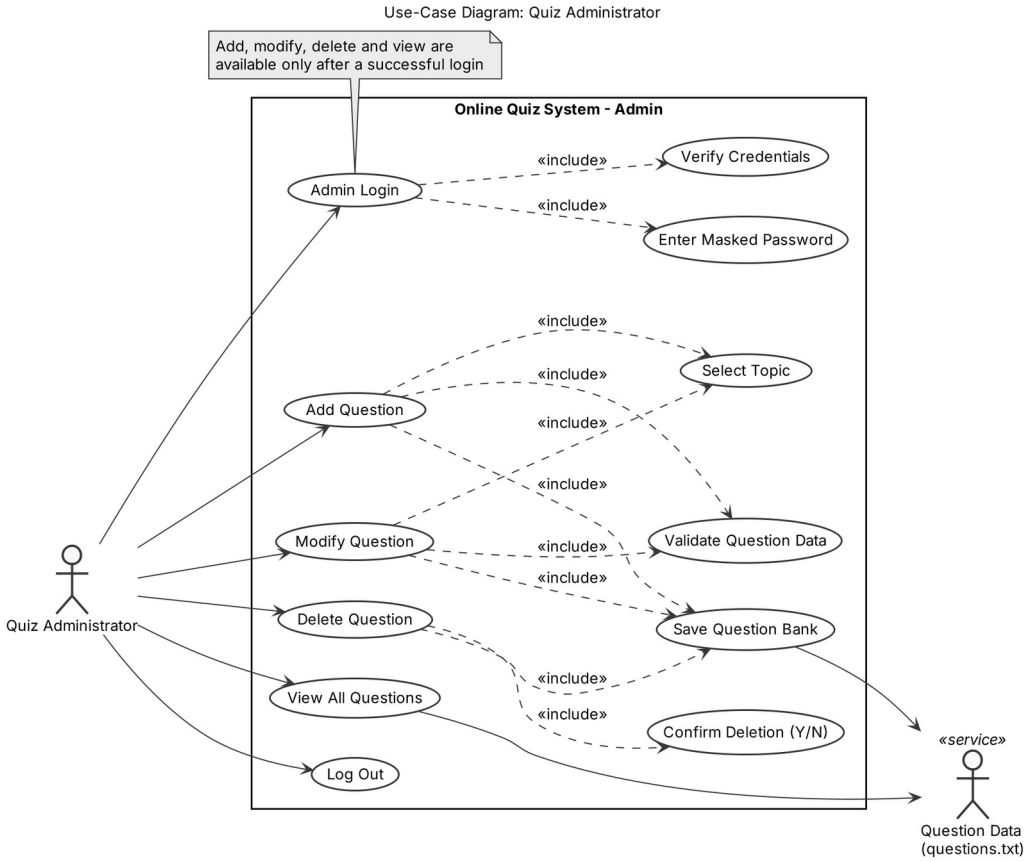


Figure 7.2 - Use-case diagram: Quiz Administrator

8. Requirements Traceability Matrix (RTM)

Section = SRS section; Design = SAD v2.0 section. Status: N = not started, P = in progress / partly verified, A = accepted (tested and passed). Results from STP v2.0 section 13.

Req ID	Requirement short	Section / Design	Module	Test case	Status (N/P/A)	Comments
OQ-F-001	Start quiz	4.1 / 3.4, 4.2	QuizController	TC-01	A	
OQ-F-002	One question at a time	4.1 / 3.4, 4.2	QuizController	TC-02	A	
OQ-F-003	Four options A-D	4.1 / 3.4, 4.2	QuizController	TC-03	A	
OQ-F-004	Accept answer	4.1 / 3.4, 4.2	QuizController	TC-04	A	
OQ-F-005	Reject invalid option	4.1 / 3.4, 4.2	QuizController	TC-05	A	
OQ-F-006	Check answer	4.2 / 3.4, 4.2	QuizController, QuizLogic	TC-06	A	
OQ-F-007	Score +1 if correct	4.2 / 3.4, 4.2	QuizLogic	TC-07	A	
OQ-F-008	Score same if wrong	4.2 / 3.4, 4.2	QuizLogic	TC-08	A	
OQ-F-009	Next question	4.1 / 3.4, 4.2	QuizController	TC-09	A	
OQ-F-010	Final score	4.2 / 3.4, 4.2	QuizLogic	TC-10	A	
OQ-F-011	Show final score	4.2 / 3.4, 4.2	QuizController	TC-11	A	
OQ-F-012	Admin add question	4.4 / 3.4, 4.2	Admin Module (main.cpp), AdminValidation	TC-12	A	
OQ-F-013	Admin modify question	4.4 / 3.4, 4.2	Admin Module (main.cpp), AdminValidation	TC-13	A	
OQ-F-014	Admin delete question	4.4 / 3.4, 4.2	Admin Module (main.cpp)	TC-14	A	
OQ-F-015	Block unauthenticated users	4.4 / 3.4, 4.2	QuizController, Admin Module (main.cpp)	TC-15	A	
OQ-F-016	Reset score on new attempt	4.1 / 3.4, 4.2	QuizController, QuizLogic	TC-16	A	
OQ-F-017	Topic question bank	4.3 / 3.4, 4.2	QuestionLoader, Question bank	TC-27	A	
OQ-F-018	Question record format	4.3 / 3.4, 4.2	QuestionLoader, AdminValidation	TC-28	A	
OQ-F-019	Topic selection	4.3 / 3.4, 4.2	QuizController, QuestionLoader	TC-29	A	
OQ-F-020	Five random questions	4.3 / 3.4, 4.2	QuizController	TC-30	A	
OQ-F-021	Avoid repeating previous set	4.3 / 3.4, 4.2	QuizController	TC-31	A	
OQ-F-022	Result menu	4.1 / 3.4, 4.2	QuizController	TC-32	A	
OQ-NF-001	Response <= 1 s	5 / 3.4, 3.7	QuizController	TC-17	A	
OQ-NF-002	Score reliability	5 / 3.4, 3.7	QuizLogic	TC-18	A	
OQ-NF-003	Clear messages	5 / 3.4, 3.7	QuizController, Admin Module (main.cpp)	TC-19	P	Manual test pending
OQ-NF-004	No exposure of admin data	5 / 3.4, 3.7	QuizController	TC-20	A	

Req ID	Requirement short	Section / Design	Module	Test case	Status (N/P/A)	Comments
OQ-NF-005	Portability	5 / 3.4, 3.7	All modules	TC-21	P	Windows run pending
OQ-SR-001	Login before management	5.1.2 / 3.9	Admin Module (main.cpp)	TC-22	A	
OQ-SR-002	Reject wrong credentials	5.1.2 / 3.9	Admin Module (main.cpp), AdminValidation	TC-23	A	Test account, see 5.1.3
OQ-SR-003	Validate admin input	5.1.2 / 3.9	Admin Module (main.cpp), AdminValidation	TC-24	A	
OQ-SR-004	No direct admin access	5.1.2 / 3.9	QuizController	TC-25	A	
OQ-SR-005	Hide credentials	5.1.2 / 3.9	Admin Module (main.cpp)	TC-26	P	"*" masking: manual check pending